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3 Automation In
Inventory Management

1 CITY-SCALE SURVEILLANCE Being Powered By AI

➤ **Implementation of surveillance and security sector technologies must be robust, reliable and address real customer challenges**

China's controversial deployment of an artificial intelligence (AI)-powered mass surveillance system to bring control to its cities has been stirring a lot of discussions lately. The general concern is the extent to which this will affect the privacy and freedom of the citizens and how well it aligns with democratic ground rules. However, it also showcases the power of AI in surveillance.

Chinese administrators can identify any citizen on the road through facial recognition or predict their motives through behavioural analysis. The system has helped them catch many crime-offenders from massive crowds. Essentially, it has hinted that a surveillance network, if scalable enough, can actually take the responsibility of securing a full-scale city.

Smart cameras powered by computer vision are being considered as the main source of video surveillance for smart cities. In the surveillance industry today, machine learning is mostly used



for video analytics. The technology has become an important component in a variety of applications and use-cases. Software engineers are working towards including it in many different environments based on compelling use-cases like city-level security. However, implementations have barely begun.

Johan Paulsson, chief technology officer, Axis Communications, says, "The surveillance and security sector is often tempted to over-promise solutions with new technologies. This has been significantly observed in the case of AI in video analytics, specifically when it comes to deep learning. Implementation of such technologies have to be robust, reliable and address real customer challenges."

Deep learning happens in two steps: a model training step and an execution phase. Deep learning platforms crunch through a lot of historical data to learn and understand execution patterns. This requires powerful processing systems, necessitating investments in